

Chapter 1

Describing Data

Probability and Statistics for Science and Engineering
with Examples in R

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(1) Display of Data by Graphs and Tables

(A) Frequency Distributions

Basic Steps

1. Find the minimum and the maximum values in the data set.
2. Class intervals: intervals of equal length that cover the range between the min and max without overlapping
eg) min 0, max 100:
 $[0, 10), [10, 20), \dots, [90, 100]$
3. Frequency: #observations in the data that belong to each interval
 $f_1, f_2, \dots$
4. Relative frequency: class frequency/total #observations
 $f_1/n, f_2/n, \dots$

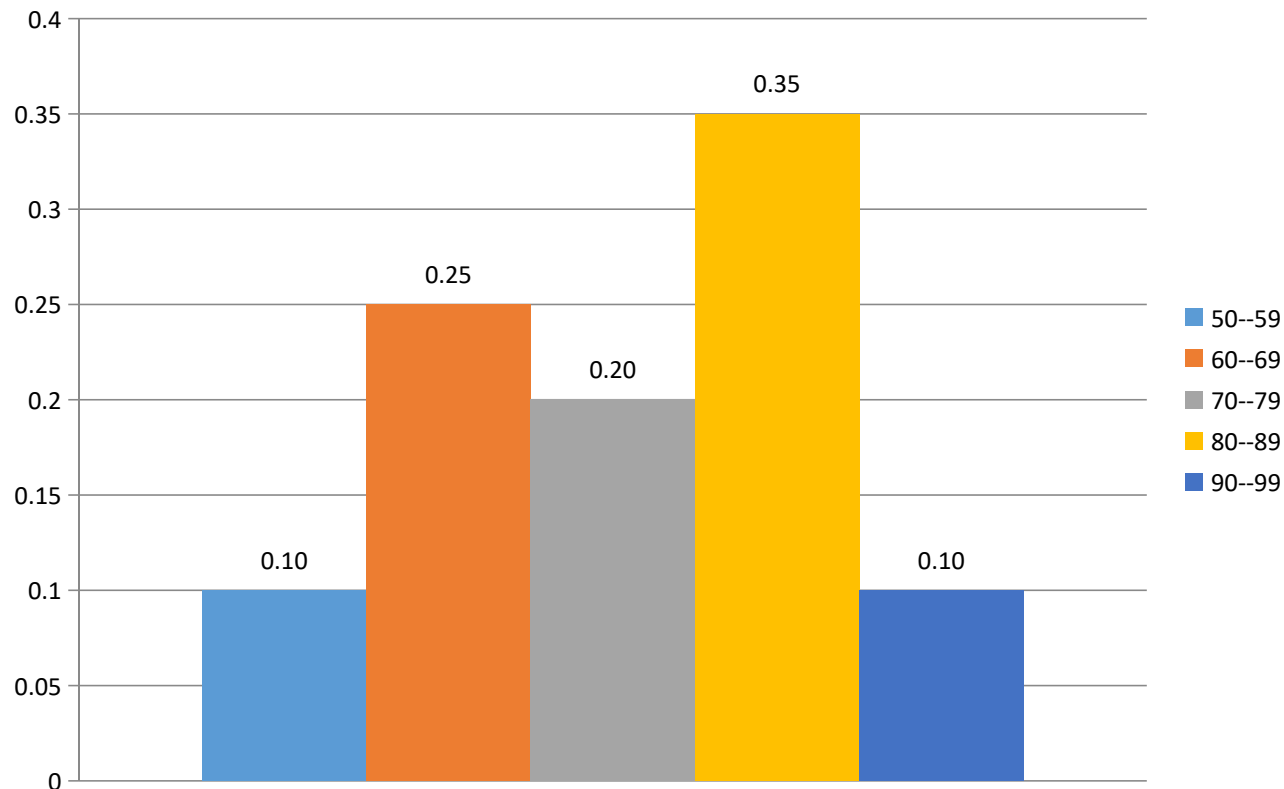
Example 1.1: Midterm Scores

69 84 52 93 81 74 89 85 88 63 87 64 67 72 74 55 82 91 68 77

Class Interval	Tally	Frequency	Rel. Freq.
50-59		2	$2/20=0.10$
60-69		5	$5/20=0.25$
70-79		4	$4/20=0.20$
80-89	-	7	$7/20=0.35$
90-99		2	$2/20=0.10$
Total		20	1.00

(B) Histogram: A pictorial representation of a frequency distribution

- Midterm Scores (continued)



$$\text{Height} = \frac{\text{Relative Frequency}}{\text{Width of the interval}} =$$

$$\frac{0.10}{10} = 0.010 \text{ for } [50, 60)$$

$$\frac{0.25}{10} = 0.025 \text{ for } [60, 70)$$

$$\frac{0.20}{10} = 0.020 \text{ for } [70, 80)$$

$$\frac{0.35}{10} = 0.035 \text{ for } [80, 90)$$

$$\frac{0.10}{10} = 0.010 \text{ for } [90, 100)$$

Shapes of Distributions

(a) Symmetric

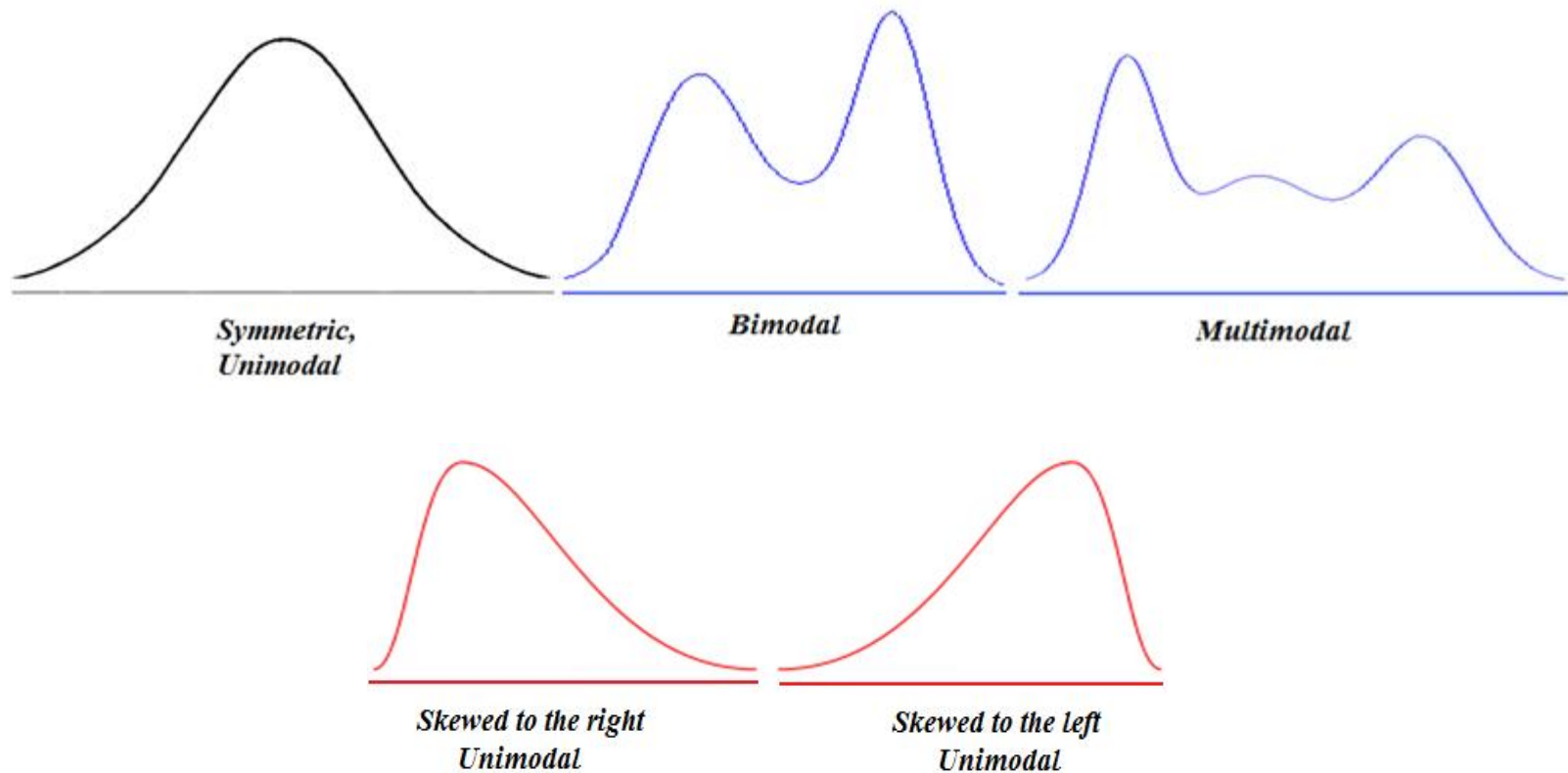
(b) Unimodal: *single peak*

(c) Bimodal: *two peaks*

(d) Uniform: The possible values appear with *equal frequency*

(e) Skewed: One side of the distribution is stretched out longer than the other side

Shapes of Distributions



Drawing a Histogram Using R

Example 1.2: Midterm exam scores

30, 34, 38, 44, 45, 46, 47, 48, 50, 50, 51, 52, 53, 53, 53, 54, 55,
55, 55, 56, 56, 57, 57, 58, 58, 59, 59, 60, 60, 60, 60, 61, 61, 62,
62, 62, 62, 63, 63, 63, 63, 63, 63, 63, 64, 64, 65, 65, 65, 65, 65,
66, 66, 67, 67, 67, 68, 68, 68, 68, 68, 69, 69, 69, 69, 69, 69, 69,
70, 70, 70, 70, 70, 70, 71, 71, 71, 72, 72, 73, 73, 73, 73, 73, 73,
73, 73, 73, 74, 74, 74, 75, 75, 75, 76, 76, 76, 76, 76, 76, 77, 77,
77, 77, 77, 78, 78, 78, 78, 78, 79, 79, 79, 80, 80, 80, 80, 81, 81,
81, 81, 82, 82, 82, 82, 82, 83, 83, 83, 83, 84, 84, 84, 84, 84, 84,
84, 84, 85, 85, 86, 86, 87, 87, 87, 87, 88, 88, 88, 88, 88, 88, 89,
89, 89, 89, 89, 90, 90, 90, 91, 92, 93, 93, 94, 94, 94, 94, 95, 95,
95, 95, 96, 96, 96, 97, 97, 98, 98, 99, 100

Drawing a Histogram Using R

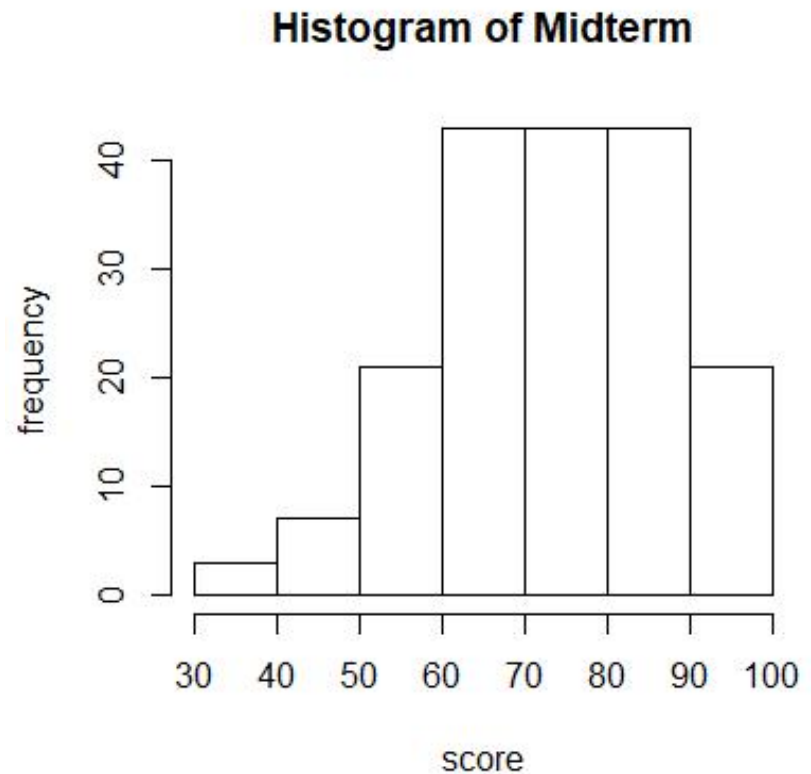
```
>midterm=read.csv("filename.csv")
```

```
>midterm=read.csv("c:\\Users\\***\\filename.csv")
```

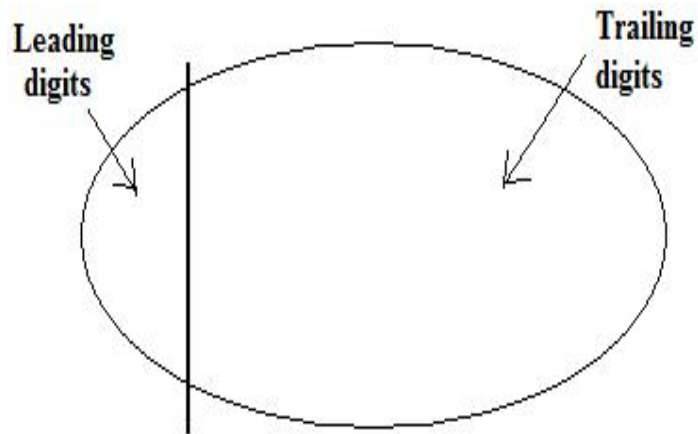
or enter the data on R as

```
>midterm=c(30, 34, 38, ..., 100)
```

```
>hist(midterm)
```



(C) Stem-and-Leaf Plot



- List one or more leading digits for the stem values.
- The trailing digit(s) become the leaves.
- Arrange the trailing digits in each row so they are in increasing order.

Example 1.3: Exam scores of 26 students

55 61 94 94 69 77 68 54 85 77 92 92 81 73 69 81 75
84 70 81 81 89 59 72 82 62

5		549
6		19892
7		773502
8		51141192
9		4422

→

5		459
6		12899
7		023577
8		11112459
9		2244

- In R,

```
>a=c(55,61,94,94,69,77,68,54,85,77,92,92,81,73,69,81,75,84,70,81,81,  
89,59,72,82,62)
```

```
>stem(a)
```

(D) Dot Diagram

Exam score data from Example 1.3

55 61 94 94 69 77 68 54 85 77 92 92 81
73 69 81 75 84 70 81 81 89 59 72 82 62



Example 1.4: Heights of students (in inches)

67.2 65.0 72.5 71.1 69.1 69.0 70.2 68.2 68.5 71.3
67.5 68.6 73.1 71.3 69.4 65.5 69.5 70.8 70.0 69.2

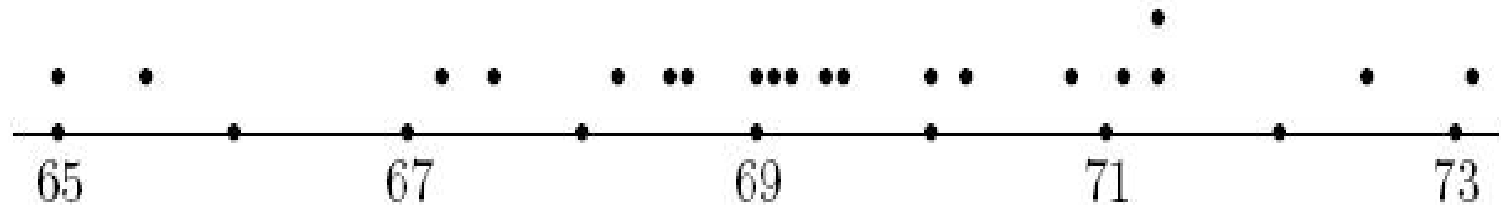
65		05
66		
67		25
68		256
69		10452
70		280
71		133
72		5
73		1

→

65		05
66		
67		25
68		256
69		01245
70		028
71		133
72		5
73		1

Dot Diagram (height data)

67.2 65.0 72.5 71.1 69.1 69.0 70.2 68.2 68.5 71.3
67.5 68.6 73.1 71.3 69.4 65.5 69.5 70.8 70.0 69.2



Side-by-Side Stem-and-Leaf Plots

Example 1.5 The same examination given to two classes

Class A: 78 80 60 74 85 100 51 60 40 67 100 90 58 40 89 100

Class B: 42 76 37 57 93 60 55 47 51 95 81 53 52 65 95

Class A		Class B
	3	7
00	4	27
18	5	12357
007	6	05
48	7	6
059	8	1
0	9	355
000	10	

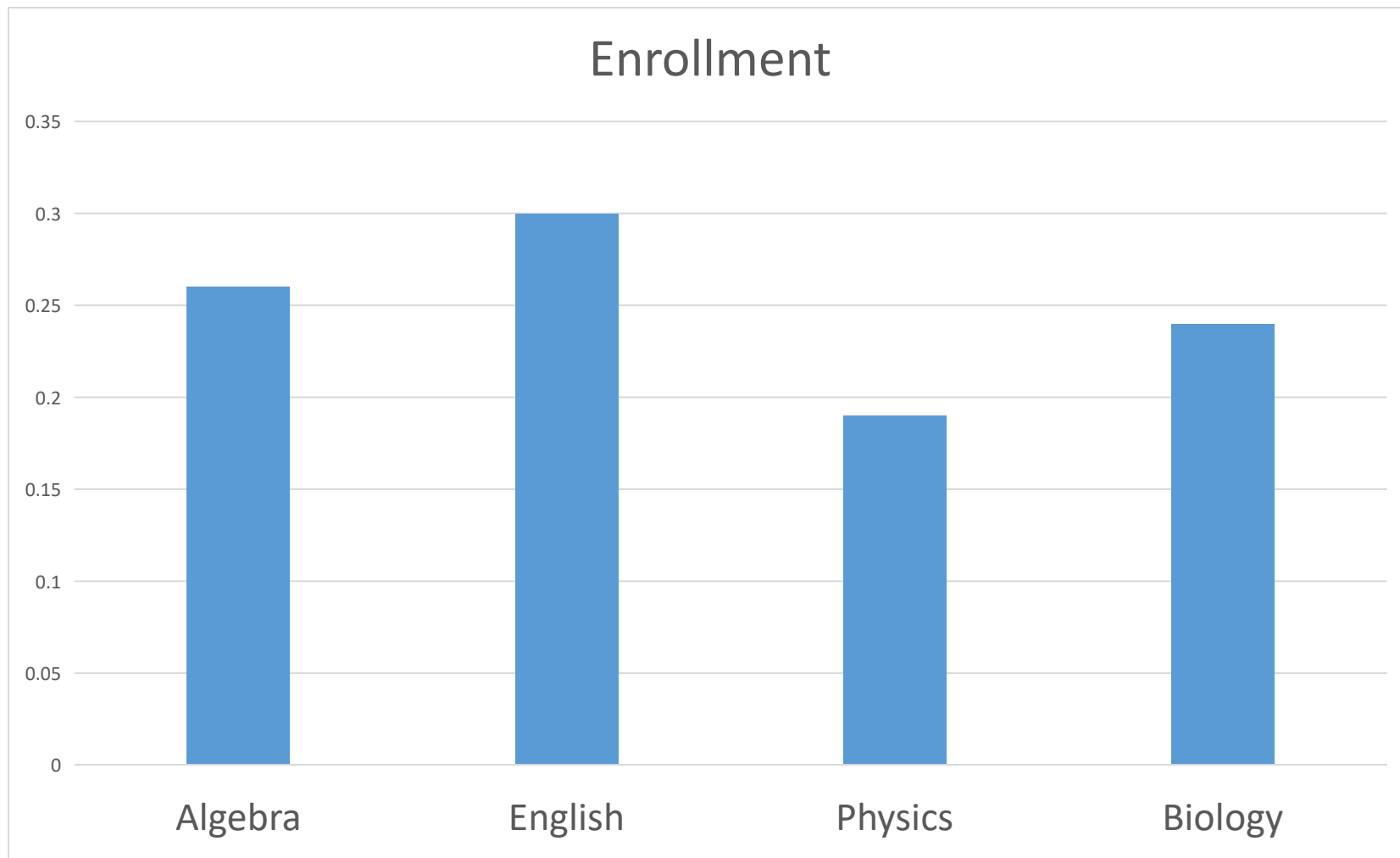
Qualitative (Categorical) Data

Example 1.6: Frequency distribution of enrollment in a high school

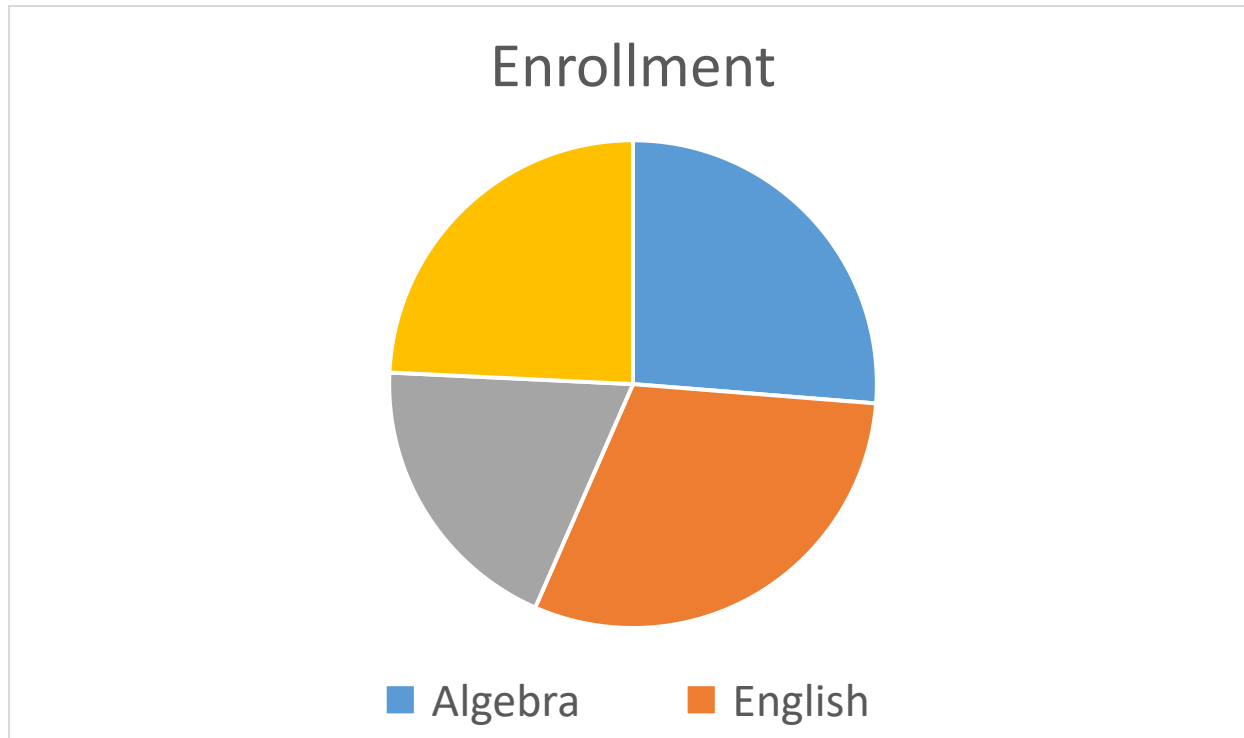
Class	Frequency	Relative Frequency
Algebra	26	0.26
English	30	0.30
Physics	19	0.19
Biology	24	0.24
Total	99	0.99

Possible round-off error

Bar Graph



Pie Chart



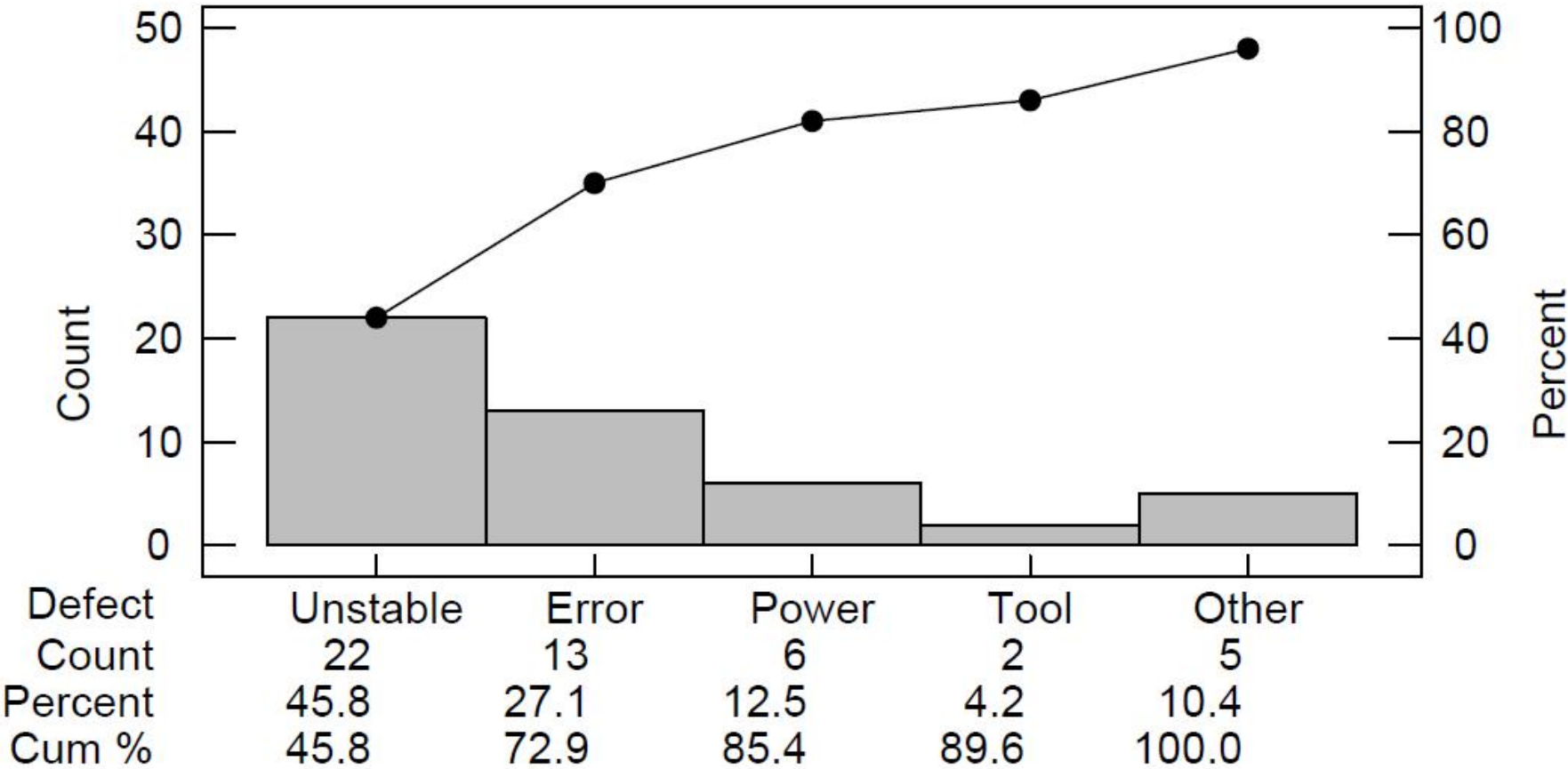
In R,

```
> slices=c(26,30,19,24)
```

```
> lbls=c("Algebra","English","Physics","Biology")
```

```
> pie(slices, labels=lbls, main="Enrollment")
```

Pareto Diagram



(E) Time Plots

- **Time plot (Time series plot):** a plot of observations against time, or the order in which they were obtained.

(a) Trends: increasing or decreasing, changes in location of the center, changes in variation

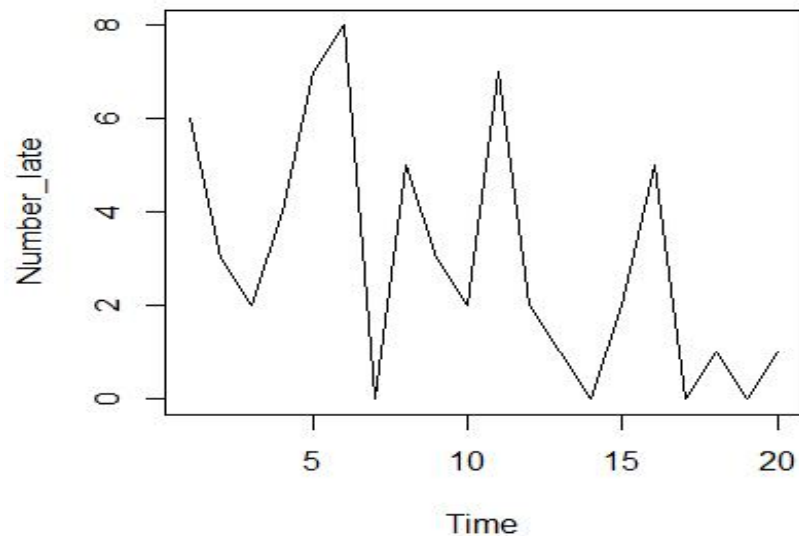
(b) Seasonal variation or cycles: fairly regular increasing or decreasing movements

In R,

```
>ts.plot
```

Example 5: Number of workers late in arriving at work

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	6	3	2	4	7
Week 2	8	0	5	3	2
Week 3	7	2	1	0	2
Week 4	5	0	1	0	1



(2) Measures of Central Tendency

Data set: $x_1, x_2, \dots, x_n$

Mean, median, mode

a) **Mode**: The number that occurs most often

b) **Mean** (average)

eg) The average height of the male adults in the USA $\Rightarrow$ Sample

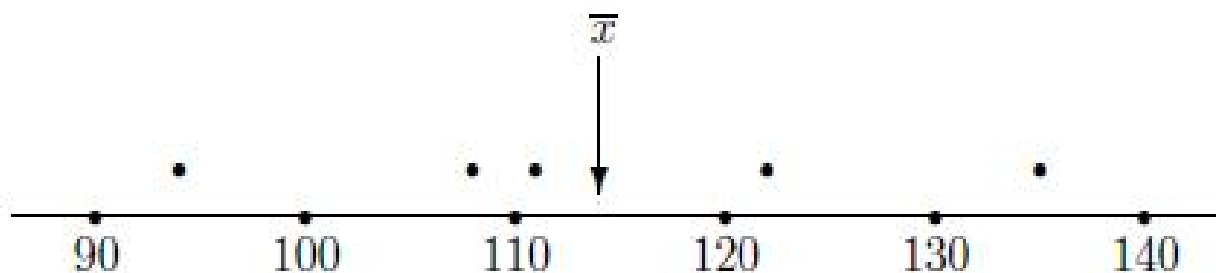
◆ Sample mean:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Example 1.7: Weights of five 7th grade girl students (in pounds)

122 94 135 111 108

$$\bar{x} = \frac{122 + 94 + 135 + 111 + 108}{5} = \frac{570}{5} = 114$$



$\bar{x}$ represents the average value of the observations in a sample. It estimates the population mean (average of all the values in the population) using $\bar{x}$.

◆ Population mean: μ

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

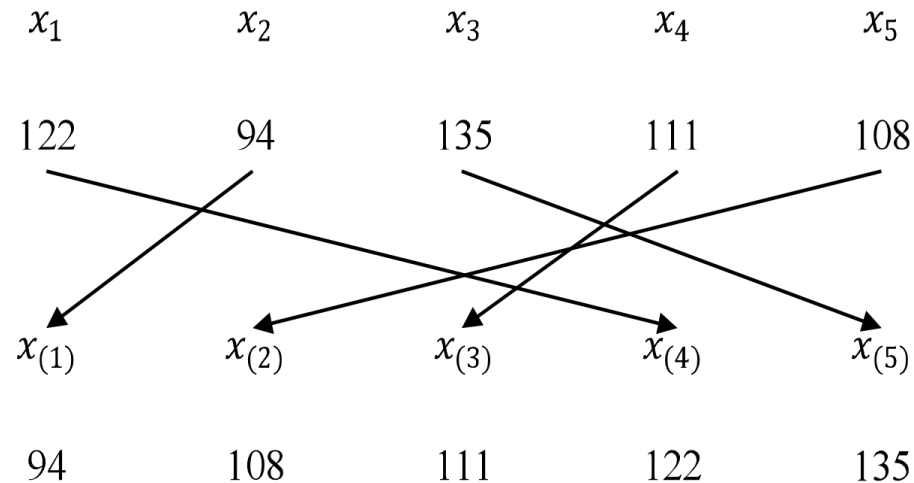
Sample median: middlemost value of the ordered data set

Algorithm for obtaining median

a) Order data

$$x_1, x_2, \dots, x_n \rightarrow x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$$

From Example 1.7,



b) i) If n is odd, take the middle value.

eg) $n = 5$: 94 108 111 122 135

Sample median:

$$\tilde{x} = x_{\left(\frac{n+1}{2}\right)} = x_{\left(\frac{5+1}{2}\right)} = x_{(3)} = 111$$

ii) If n is even, take the average of the two middle values.

Example 1.8: The number of days that the first six heart transplant patients at Stanford survived after their operations:

$$\begin{array}{cccccc}
 15 & 3 & 46 & 623 & 126 & 64 \\
 \rightarrow 3 & 15 & 46 & 64 & 126 & 623 \\
 & & \frac{46 + 64}{2} = 55
 \end{array}$$

Sample median:

$$\tilde{x} = \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2} = \frac{x_{(\frac{6}{2})} + x_{(\frac{6}{2}+1)}}{2} = \frac{x_{(3)} + x_{(4)}}{2} = \frac{46 + 64}{2} = 55$$

Sample mean:

$$\frac{15 + 3 + 46 + 623 + 126 + 64}{6} = 146.2 \text{ days}$$

Only one patient lived longer than 146.2 days.

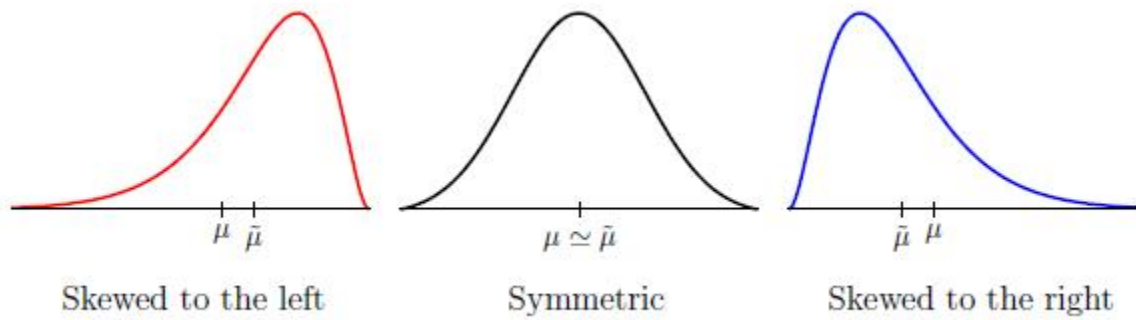
Median is a better indicator in this case.

Sample mean: greatly influenced by outlying values

Skewness

- Where the mean and median fall in a frequency distribution tells us something about the skewness of the data
- If the mean and median have the same value, then the data are not skewed.
- If the mean is greater than the median, the data are positively skewed (skewed to the right).
- If the mean is less than the median, the data are negatively skewed (skewed to the left).

Skewness



Comparison of mean, median and mode

Example 1.9: Professional baseball players' annual income

Income (in \$1,000)	Number of players
30	180
50	100
100	50
500	20
1,000	10
2,000	15
3,000	15
5,000	10
Total	400

Mean:

$$\bar{x} = \frac{180(30,000) + 100(50,000) + \dots + 10(5,000,000)}{400}$$
$$= \$401,000$$

Median:

$$\tilde{x} = \frac{x_{(200)} + x_{(201)}}{2} = \frac{50,000 + 50,000}{2} = \$50,000$$

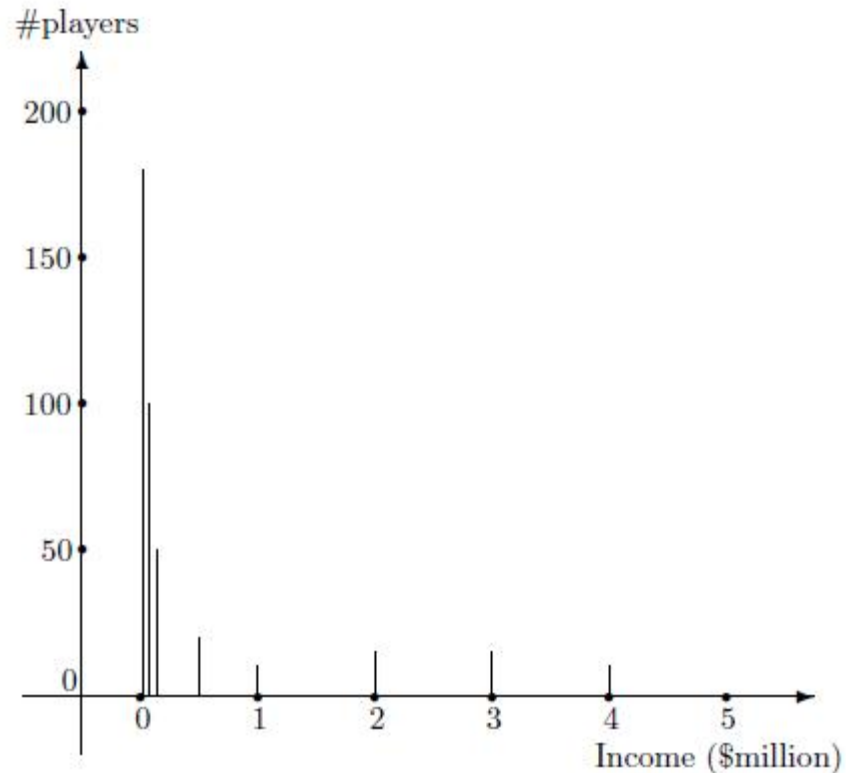
Mode:

\$30,000

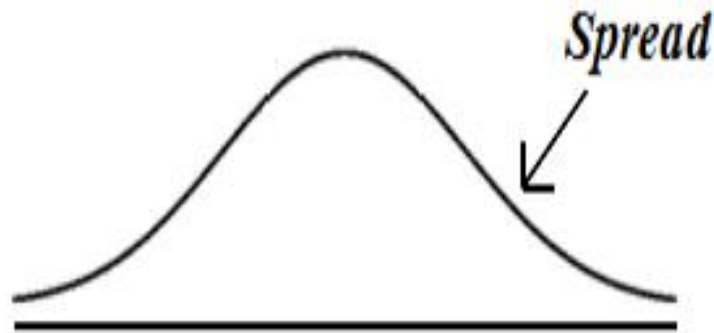
In R,

`>mean(a)`

`>median(a)`



(3) Measures of Variation



A measure of location cannot give a complete summary of data.

- Sample deviation:

Deviation: $x - \bar{x}$

i -th deviation: $x_i - \bar{x}$

Sample deviation:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} = n\bar{x} - n\bar{x} = 0$$

- Absolute deviation: $|x_1 - \bar{x}|, \dots, |x_n - \bar{x}|$
leads to theoretical difficulties
- Squared deviation: $(x_1 - \bar{x})^2, \dots, (x_n - \bar{x})^2$
has nice statistical properties

Sample variance & standard deviation

- Sample variance:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

- Sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

- Divisor: $n - 1$ instead of n
- because the x_i 's tend to be closer to $\bar{x}$ than μ
- Using n tends to underestimate σ^2 .

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2 / n}{n - 1}$$

(hand-held calculator formula)

Example 1.10: 6 12 6 6 4 8

Observation	x_i	x_i^2
1	6	36
2	12	144
3	6	36
4	6	36
5	4	16
6	8	64
Total	$\sum x_i = 42$	$\sum x_i^2 = 332$

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2 / n}{n-1} = \frac{332 - 42^2 / 6}{6-1} = 7.6$$

$$s = \sqrt{7.6} = 2.76$$

- In R,

```
>var(a)
```

```
>sd(a)
```

or

```
>sqrt(var(a))
```

Quartiles & Percentiles

Sample $100p$ -th percentile: value such that at least $100p\%$ of the observations are at or below this value; and at least $100(1-p)\%$ are at or above this value

Calculating the sample $100p$ -th percentile

1. Order the n observations from smallest to largest
2. Find np
3. If np is not an integer, round it up to the next integer and find the corresponding ordered value.

eg) $np = 3.5 \uparrow 4 \rightarrow x_{(4)}$

If np is an integer, say, k , calculate $\frac{x_{(k)} + x_{(k+1)}}{2}$.

- Quartiles: The points for division into quarters

First quartile: $Q_1 = 25^{\text{th}}$ percentile

Second quartile: $Q_2 = 50^{\text{th}}$ percentile = median

Third quartile: $Q_3 = 75^{\text{th}}$ percentile

Boxplots



Sample range: $x_{(n)} - x_{(1)} = \text{max} - \text{min}$

Interquartile range (IQR): $Q_3 - Q_1$ (length of the middle half)

Outlier: an observation which is less than $Q_1 - 1.5(\text{IQR})$ or greater than $Q_3 + 1.5(\text{IQR})$

Example 1.11: Final exam scores

81 57 85 84 99 90 69 76 76 83

Order: 57 69 76 76 81 83 84 85 90 99

$$n = 10. \quad \tilde{x} = Q_2 = \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2} = \frac{x_{(5)} + x_{(6)}}{2} = \frac{81 + 83}{2} = 82$$
$$np = 10(0.5) = 5$$

$$Q_1: np = 10(0.25) = 2.5 \uparrow 3 \rightarrow x_{(3)} = 76$$

$$Q_3: np = 10(0.75) = 7.5 \uparrow 8 \rightarrow x_{(8)} = 85$$

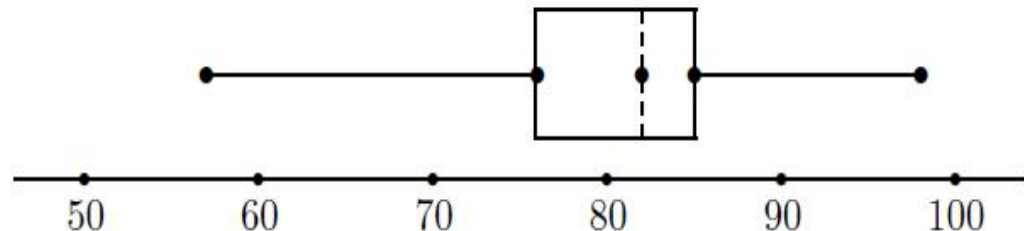
$$\text{min: } 57, \text{ max: } 99 \quad \text{range: } 99 - 57 = 42$$

$$\text{IQR} = Q_3 - Q_1 = 85 - 76 = 9$$

$$Q_1 - 1.5(\text{IQR}) = 76 - 1.5(9) = 62.5$$

$$Q_3 + 1.5(\text{IQR}) = 85 + 1.5(9) = 98.5$$

Outliers: 57 & 99



Example 1.12: 78 99 47 53 71 69 60 57 45 88 59

Order: 45 47 53 57 59 60 69 71 78 88 99

$$n = 11. \quad \tilde{x} = Q_2 = x_{\left(\frac{n+1}{2}\right)} = x_{\left(\frac{12}{2}\right)} = x_{(6)} = 60$$

$$np = 11(0.5) = 5.5 \uparrow 6$$

$$Q_1: np = 11(0.25) = 2.75 \uparrow 3 \rightarrow x_{(3)} = 53$$

$$Q_3: np = 11(0.75) = 8.25 \uparrow 9 \rightarrow x_{(9)} = 78$$

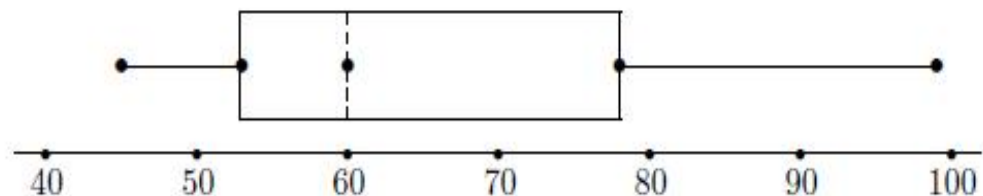
$$\text{min: } 45, \text{ max: } 99 \quad \text{range: } 99 - 45 = 54$$

$$\text{IQR} = Q_3 - Q_1 = 78 - 53 = 25$$

$$Q_1 - 1.5(\text{IQR}) = 53 - 1.5(25) = 15.5$$

$$Q_3 + 1.5(\text{IQR}) = 78 + 1.5(25) = 115.5$$

No outliers

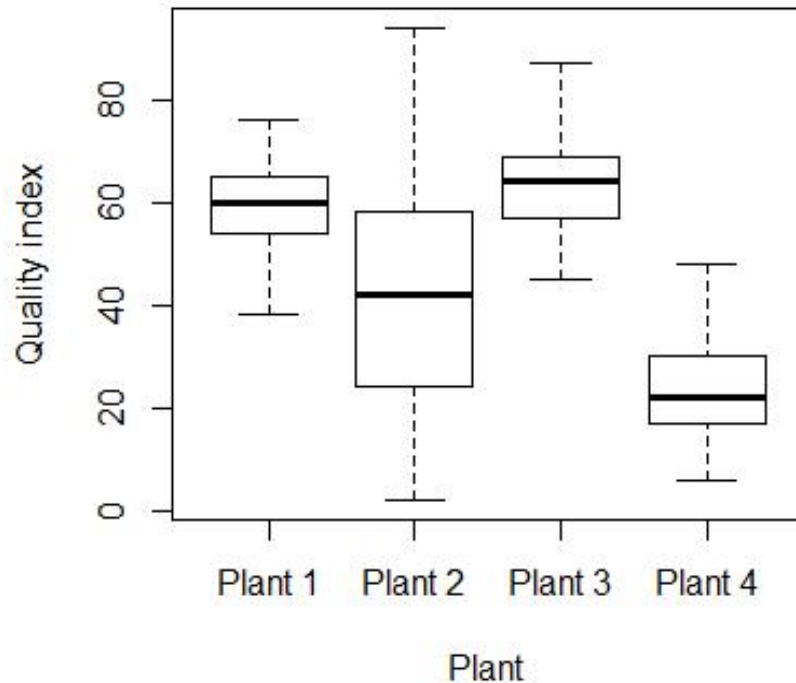


90th percentile:

$$np = 11(0.9) = 9.9 \uparrow 10 \rightarrow x_{(10)} = 88$$

Multiple Boxplots

(Reveal differences and similarities)



- Plant 2 needs to reduce its variability.
- Both Plants 2 and 4 need to improve their quality level.

Using R

- The $100p$ -th percentile of data “a” can be obtained as
`>quantile(a, p)`
- More than one percentile, say n percentiles, can be obtained as
`>quantile(a, c(p1, p2, ...,pn))`
- For the data in Example 1.11, the quartiles can be obtained as
`>a=c(81,57,85,84,99,90,69,76,76,83)`
`>quantile(a, c(0.25, 0.50, 0.75))`

25%	50%	75%
76.00	82.00	84.75

`>min(a)`
`>max(a)`
`>IQR(a)`
`>boxplot(a)`

Using R

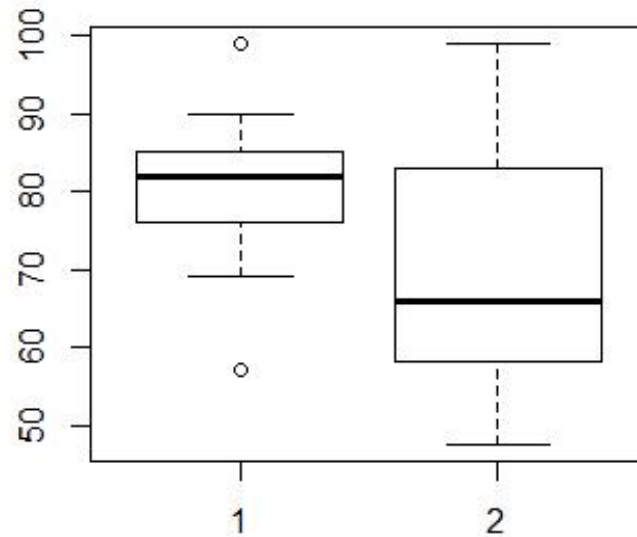
- A side-by-side boxplot of a and b can be obtained by

```
>boxplot(a, b)
```

- For Example 1.12,

```
>b=c(78,99,47,53,71,69,60,57,45,88,59)
```

```
>boxplot(a, b)
```



(4) Relationship Between Two Variables

n pairs of observations:

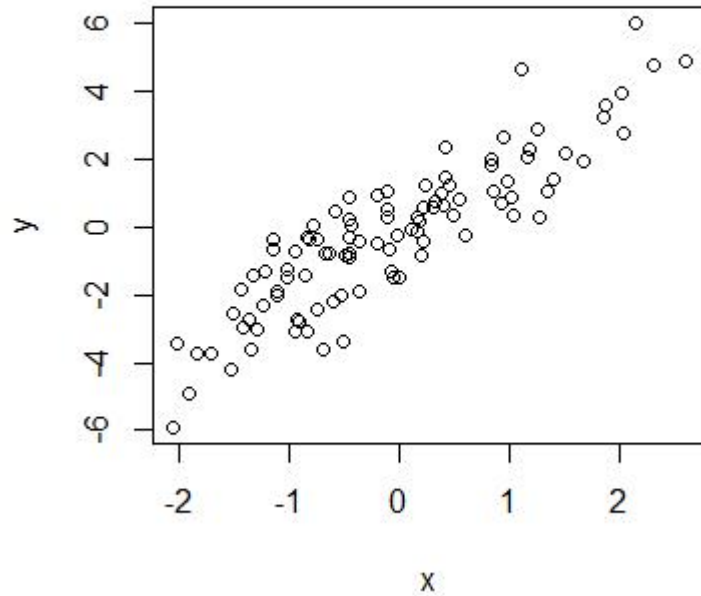
$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Covariance: measure of relationship between two variables

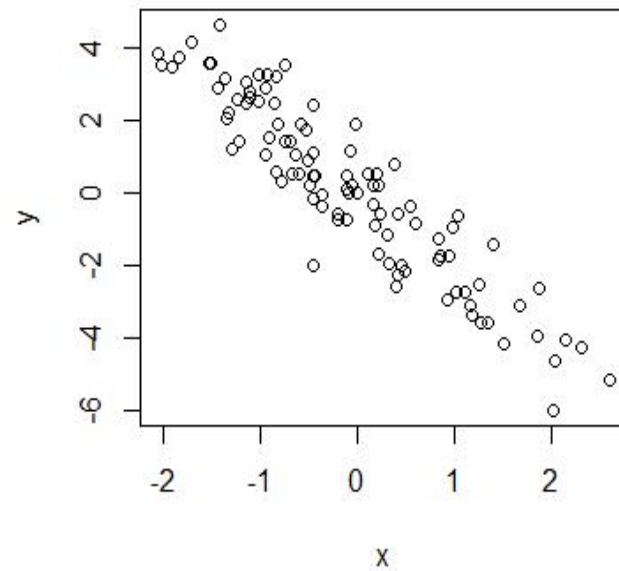
$$\begin{aligned} \diamond \quad \text{Cov}(x, y) &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} \\ &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{n-1} \end{aligned}$$

Covariance

Positive Covariance



Negative Covariance



Example 1.13: Initial speed (mph) & stopping distance (feet)

Initial speed	11	22	32	41	51
Stopping distance	8.2	32.8	82.0	144.4	236.2

$$\bar{x} = 31.4 \text{ and } \bar{y} = 100.72$$

Example 1.13 (continued)

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	11	8.2	-20.4	-92.52	1887.41
2	22	32.8	-9.4	-67.92	638.45
3	32	82.0	0.6	-18.72	-11.23
4	41	144.4	9.6	43.68	419.33
5	51	236.2	19.6	135.48	2655.41
Total					5589.36

$$\bar{x} = 31.4 \text{ and } \bar{y} = 100.72$$

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1} = \frac{5589.36}{5 - 1} = 1397.34$$

Example 1.13 (continued)

i	x_i	y_i	$x_i y_i$
1	11	8.2	90.2
2	22	32.8	721.6
3	32	82.0	2624.0
4	41	144.4	5920.4
5	51	236.2	12046.2
Total			21402.4

$$\bar{x} = 31.4 \text{ and } \bar{y} = 100.72$$

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{n - 1} = \frac{21402.4 - 5(31.4)(100.72)}{5 - 1} = 1397.34$$

Correlation Coefficient

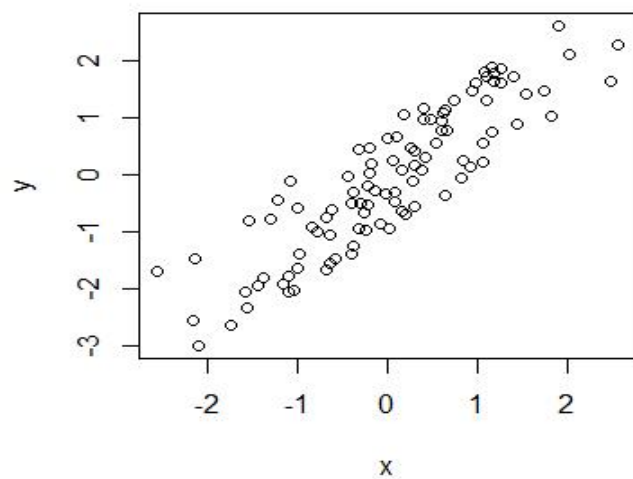
- Covariance is affected by the magnitude of the variables.
- Correlation coefficient: measure of relationship with a standard unit
- Sample correlation coefficient:

$$r = \frac{Cov(x, y)}{s_x s_y}$$

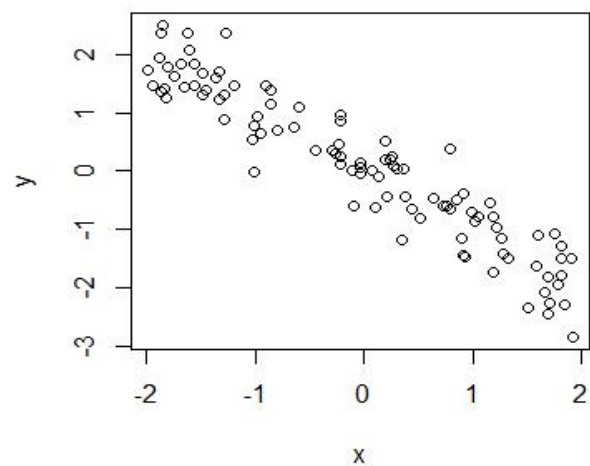
where s_x and s_y are sample standard deviations of x and y , respectively

$$-1 \leq r \leq 1$$

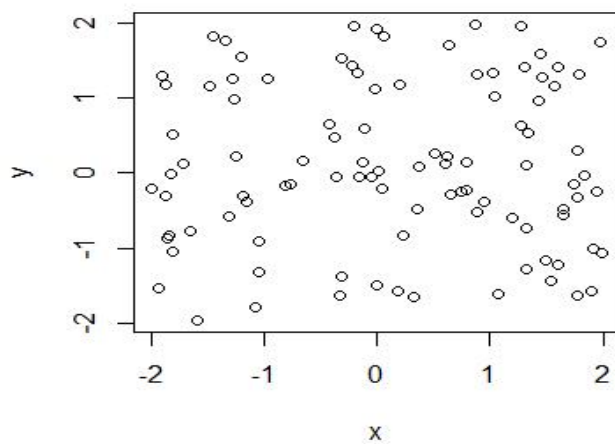
(a) Correlation Near 1



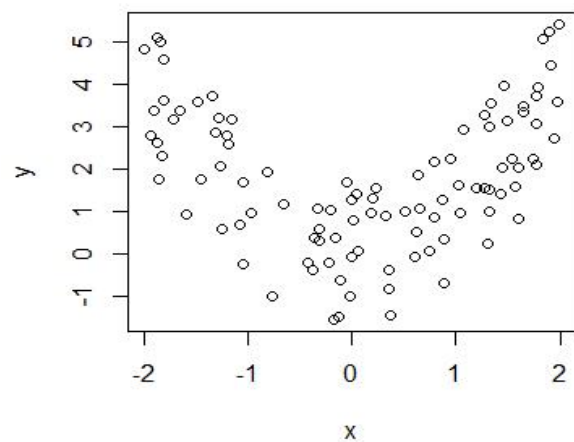
(b) Correlation Near -1



(c) Correlation Near 0



(d) Correlation Near 0



Example 1.13 (continued)

Initial speed	11	22	32	41	51
Stopping distance	8.2	32.8	82.0	144.4	236.2

$$\bar{x} = 31.4 \text{ and } \bar{y} = 100.72$$

$$s_x = \sqrt{\frac{\sum_{i=1}^n x_i^2 - n(\bar{x})^2}{n-1}} = \sqrt{\frac{5,911 - 5(31.4^2)}{5-1}} = 15.66$$

$$s_y = \sqrt{\frac{\sum_{i=1}^n y_i^2 - n(\bar{y})^2}{n-1}} = \sqrt{\frac{84,508.88 - 5(100.72^2)}{5-1}} = 91.91$$

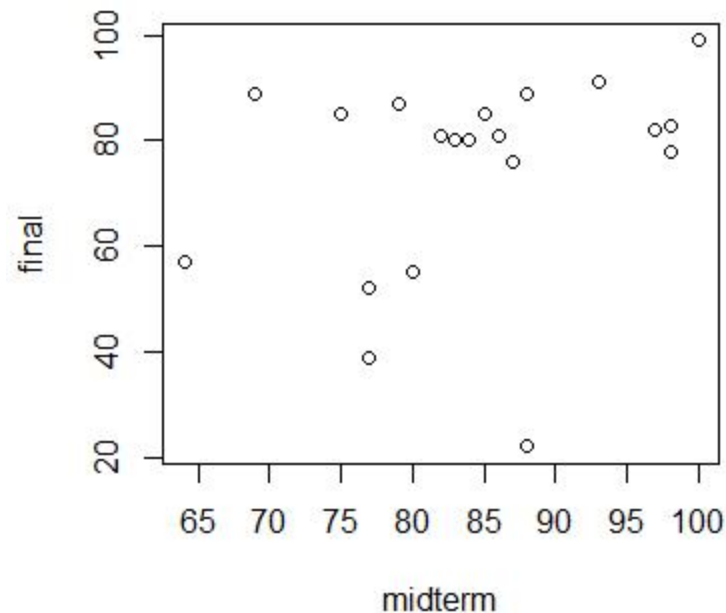
$$r = \frac{Cov(x, y)}{s_x s_y} = \frac{1,397.34}{(15.66)(91.91)} = 0.97$$

Example 1.15: Exam scores

Student	1	2	3	4	5	6	7	8	9	10
Midterm	82	80	88	64	69	79	97	75	98	88
Final	81	55	22	57	89	87	82	85	78	89

Student	11	12	13	14	15	16	17	18	19	20
Midterm	93	87	86	83	85	84	98	100	77	77
Final	91	76	81	80	85	80	83	99	39	52

Exam Scores



Example 1.15: Exam scores (continued)

- Correlation coefficient

$$\begin{aligned} Cov(x, y) &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{n - 1} = \frac{127,112 - 20(84.50)(74.55)}{20 - 1} \\ &= 59.08 \end{aligned}$$

$$s_x = \sqrt{\frac{\sum_{i=1}^n x_i^2 - n(\bar{x})^2}{n - 1}} = \sqrt{\frac{144,594 - 20(84.5^2)}{20 - 1}} = 9.70$$

$$s_y = \sqrt{\frac{\sum_{i=1}^n y_i^2 - n(\bar{y})^2}{n - 1}} = \sqrt{\frac{118,321 - 20(74.55^2)}{20 - 1}} = 19.42$$

$$r = \frac{Cov(x, y)}{s_x s_y} = \frac{59.08}{(9.70)(19.42)} = 0.31$$

Correlation Coefficient

- In R,

Scatter plot:

`>plot(x, y, main= (title of the plot) )`

Covariance:

`>cov(x, y)`

Correlation coefficient:

`>cor(x, y)`

For Example 1.15, the scatter plot is obtained by
`>plot(midterm, final, main= Exam Scores )`